OO-F-589b
June 23, 1964

SUPERSEDING
Fed. Spec. 00-F-589a
January 14, 1954

FEDERAL SPECIFICATION

FORMING MACHINE, SHEET METAL, SLIP ROLL HAND-OPERATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers manually operated bench and pedestal slip roll sheet metal forming machines suitable for forming flat sheet metal into cylindrical and curved shapes.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. Slip roll formers furnished under this specification shall be of the initial pinch type, hand operated, and of the following styles, as specified (see 6.2):

Style I.—Bench type.

Style II.—Pedestal type.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-18058—Preparation for Delivery of Machine Tools (Metal and Woodworking), Accessories and Repair Parts.

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Military Standards:

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Application for copies of publications should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of slip roll forming machines which are otherwise in accordance with the requirements of this specification.

3.2 Preproduction sample. Unless otherwise specified (see 6.2), the contractor shall furnish a complete slip roll forming machine for preproduction inspection in accordance with section 4. The sample may be a preproduction item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used or were used in production of items to fulfill the terms of the contract. Approval of a preproduction sample shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.3 Design. Slip roll forming machines furnished under this specification shall be new and of the manufacturer's latest design. The design shall be such as to provide convenient, efficient, and safe operation. All parts subject to wear shall be easily accessible for adjustment, replacement and repair.

3.3.1 *Safety devices.* The machine shall be fitted with safety devices or covering for all parts that present safety hazards. All guards shall be designed to provide access to the guarded parts. When special safety devices are required, they shall be as specified (see 6.2).

3.3.2 *Lubrication.* All bearings (except sealed for life type), mating gears, and sliding parts shall be provided with means for lubrication. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.3.3 *Interchangeability.* All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.3.4 *Material.* All parts used in the fabrication of the machine shall be new and of the quality necessary to produce a machine to meet the requirements described herein.

3.4 Construction. The machine shall be complete so that when installed it can be used for any operation for which it was designed. Construction shall be free from any characteristics or defects that will prevent the machine from passing any of the examinations and tests in section 4.

3.4.1 *Castings and forgings.* All castings and forgings shall be free from defects,

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scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with cold solders or metallic pastes be used on castings or forgings for reclaiming any parts of forming machines.

3.4.2 *Welding, brazing, or soldering.* Welding, brazing, or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.4.3 *Fastening devices.* All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.4 *Surfaces.* All surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.4.5 *Painting.* Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.4.6 *Threads.* All threaded parts shall be in the inch system and shall conform to Bureau of Standards Handbook H-28.

3.5 Performance and accuracy. This machine shall be capable of forming a cylinder true within 1/16 inch that shall not taper by more than 1/16 inch.

3.6 Style I, bench. Style I slip rolls shall be of the bench type suitable for mounting on a bench or pedestal. The rolls shall conform to the requirements of table I for the size specified (see 6.2). Style I machine shall be similar to figure 1 and shall consist essentially of a base, end frames, drive mechanism, rolls, roll adjustment, gear housing, hand crank, and locking mechanism.

TABLE I.—*Detail requirements for style I and style II slip roll forming machines*

Size number	Roll diameter	Effective working roll length	Capacity in mild steel, at least		Minimum net weight	
			Manufacturer's standard gage	Thickness	Style I	Style II
	Inches	*Inches*		*Inch*	*Pounds*	*Pounds*
0	1	6	16	0.0598	40	—
1/2	1	12	22	.0299	45	—
1	1	16	28—30	.0149—.0120	45	—
1-1/2	1-1/2	20	24—26	.0239—.0179	85	—
2-A	2	24	20	.0359	158	—
2	2	30	22	.0299	170	215
3	2	36	22	.0299	210	240
3-1/2	2	42	22—24	.0299—.0239	260	275
4	2-1/2	36	16—18	.0598—.0478	430	550
4-B	2-1/2	48	20—22	.0359—.0299	520	650
5	3	36	14—16	.0747—.0598	580	650
5-1/2	3	42	15—18	.0673—.0478	620	680
6	3	48	16—18	.0598—.0478	660	725
7	4	48	12—14	.1046—.0747	—	1675
8	4	72	16—18	.0598—.0478	—	1750

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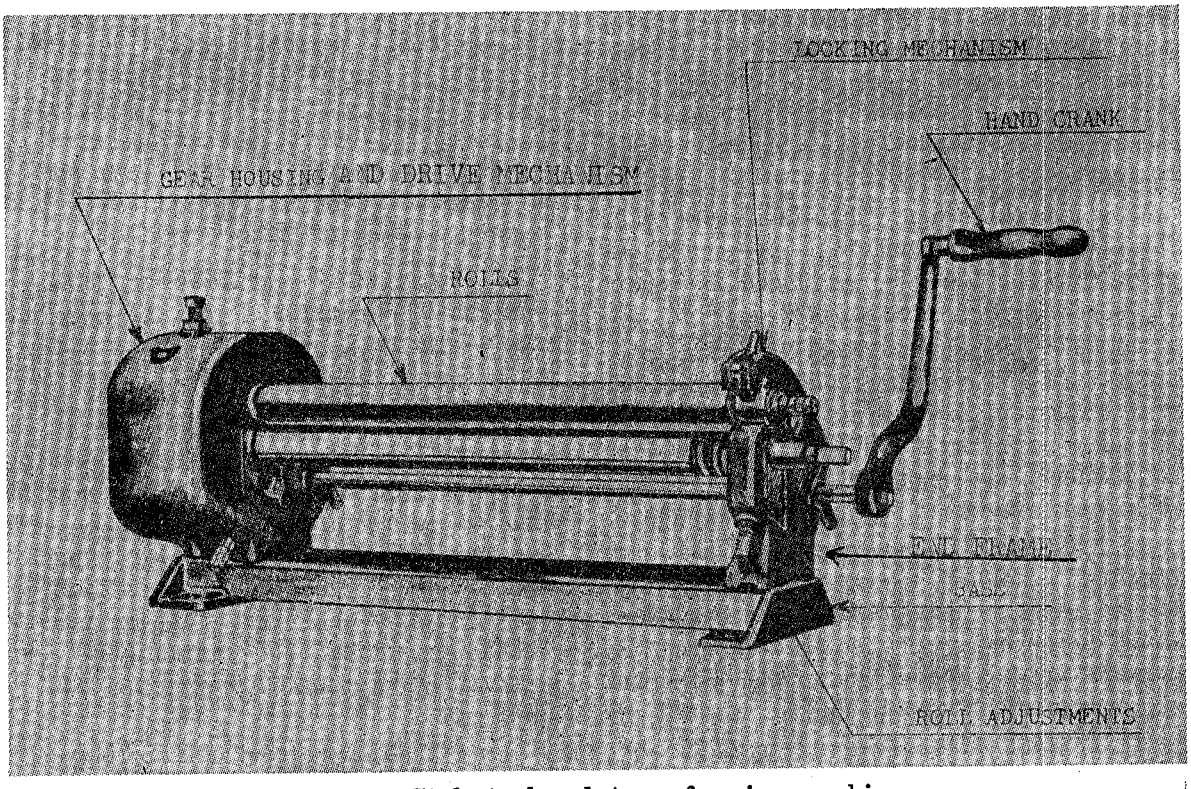


FIGURE 1.—Style 1—bench type, forming machine.

3.6.1 *Base.* The base shall be an iron alloy casting or a fabricated steel weldment of such design and weight as will support the end frames, rolls, drive mechanism and other components of the forming machine at maximum rated capacity. The base shall have four through holes for mounting.

3.6.2 *End frames.* End frames shall be an iron alloy casting or steel weldment. End frames shall house the roll bearings. The left end frame shall house the swivel yoke and gear assembly.

3.6.3 *Drive.* The forming rolls shall be geared together and arranged for operation from a handcrank. All gears shall be of steel or cast iron and with machined teeth. The gears in the train shall maintain proper mesh when the upper roll is opened or closed, and proper mesh shall also be maintained when the front rolls are in contact or separated. The rear roll gears shall be so mounted as to insure meshing in any position of the rear roll. All revolving elements of the drive shall be provided with means for lubrication.

3.6.4 *Rolls.* Machines having rolls 2-1/2 inches in diameter and smaller may be gear driven on pinch rolls only. Machines having rolls 3 inches in diameter and larger shall have all rolls gear driven. The rolls shall be made of steel. When specified (see 6.2), rolls shall be hardened to at least 45 Rockwell C and finished to 32 r.m.s. microinch.

3.6.4.1 *Roll adjustments.* The lower front roll shall be provided with vertical adjustment to obtain the desired clearance between the top and bottom feed rolls. The rear roll shall be adjustable to provide the required radius of curvature of the cylinder being formed.

3.6.4.2 *Roll bearings.* Unless otherwise specified (see 6.2), main roll bearings shall be cast iron of the sleeve type.

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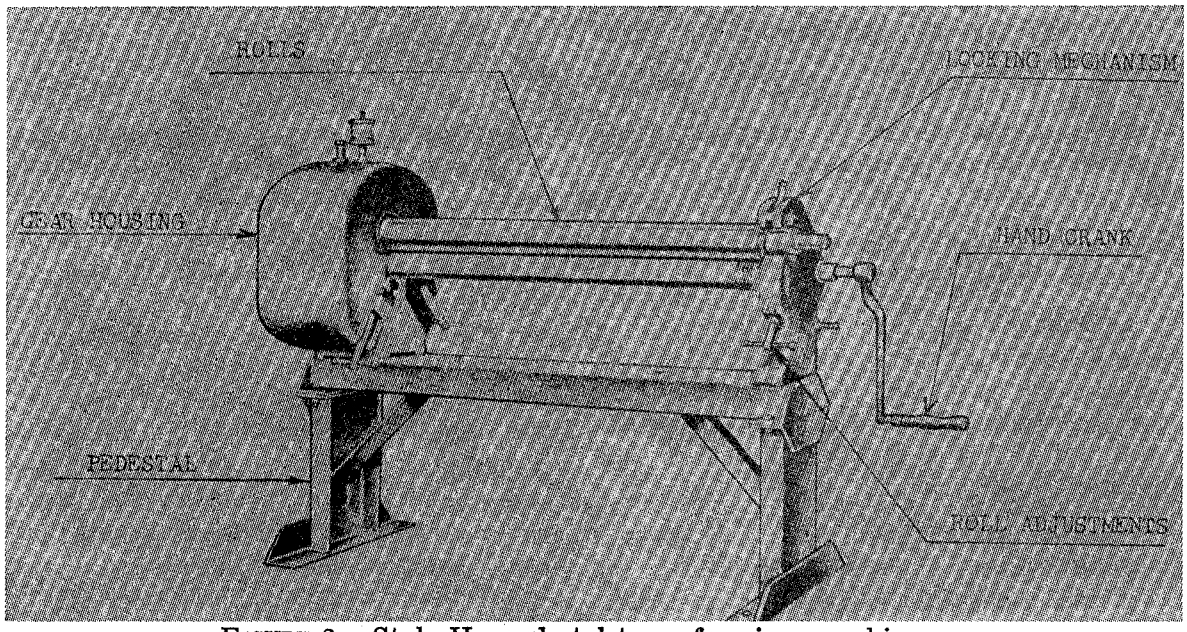


FIGURE 2.—Style II—pedestal type, forming machine.

3.6.5 *Slip roll feature.* The upper roll shall be pivoted at the gear end and so mounted that the other end may be raised or swung away from the stationary rolls in order to permit formed pipes or cylinders, encircling the upper roll to be withdrawn without distortion. Where the design is such that the top roll must be raised, it shall be provided with a lever, crank, or other device for tilting.

3.6.6 *Locking mechanism.* The right hand end of the upper roll shall be held in position with a quick acting locking mechanism which shall include a lever or trigger release allowing the upper roll to swing up or out to permit removal of a formed cylinder.

3.6.7 *Wire groove.* Unless otherwise specified (see 6.2), formers having 3-inch and 4-inch diameter rolls shall have grooves provided at the right hand end of the rear and bottom rolls to permit forming work with outside wired or rolled edges. The width of the semicircular grooves shall be 3/8, 1/2, and 5/8 inch.

3.6.8 *Reversing feature.* It shall be possible to reverse the direction of rotation of the rolls without binding or marring the work.

3.7 Style II, pedestal. Style II slip rolls shall be of the pedestal type suitable for floor mounting. The forming machine shall be similar to figure 2 and shall conform to the requirements of table I for the size specified (see 6.2.) Style II machine shall be essentially the same as style I except as follows:

3.7.1 *Pedestal or base.* The base, pedestal, or floor stand for style II forming machines shall be of cast iron or a fabricated steel weldment, designed as an integral part of the end frame or as a base on which the end frame can be mounted. Through holes for mounting shall be provided. When specified (see 6.2), the floor stand or pedestal shall be so constructed that it may be disassembled from the forming machine.

3.8 Equipment.

3.8.1 *Standard equipment.* Accessories and tools normally furnished by the manufacturer shall be provided.

3.8.2 *Optional equipment.* When specified (see 6.2), the following optional equipment shall be furnished.

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3.8.2.1 *Cone-rolling attachment.* A cone-rolling attachment shall be capable of forming cones within the capacity of the machine and shall be easily detachable from the former when not in use.

3.8.2.2 *Longitudinal groove.* The bottom and rear roll shall be provided with a longitudinal groove.

3.9 **Nameplate.** Unless otherwise specified (see 6.2), a corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each machine:

Manufacturer's Name. .....
Manufacturer's Model Number. .....
Manufacturer's Serial Number. .....
Contract Number. .....
Date of Manufacture. .....
U. S. .....

3.10 **Identification plate.** When specified (see 6.2), all machines supplied shall carry a separate identification plate of corrosion resistant metal permanently and legibly marked with the following information. Figures shall be at least 3/16 of an inch in height.

Production equipment code number (to be supplied in contract).
Activity identification (to be inscribed by the receiving activity).

3.11 **Handbooks of instructions and parts list.** Unless otherwise specified (see 6.2), two copies each of the manufacturer's standard operators manual and parts list shall be placed in a waterproof envelope marked "TECHNICAL DATA" and securely attached to each machine.

3.12 **Anchor bolt plans.** Unless otherwise specified (see 6.2), two copies of anchor bolt plans shall be furnished.

3.13 **Workmanship.** Workmanship of slip roll forming machines and accessories shall be of the quality prevailing among manufacturers, normally producing forming machines of types specified herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Preproduction inspection.** Preproduction inspection shall be applied to the preproduction sample submitted in accordance with 3.2. Failure of the preproduction sample to pass the examination or one or more preproduction tests shall be cause for rejection. Unless otherwise specified (see 6.2), preproduction inspection shall consist of the examination in 4.4 and tests in 4.5.

4.3 **Quality conformance inspection.** Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Failure of the machine to pass the examination or one or more tests shall be cause for rejection. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the following:

(*a*) Examination (see 4.4).
(*b*) Test (see 4.5).
(*c*) Examination of preservation, packaging, packing, and marking for shipment and storage (see 4.6).

4.4 **Examination.** The machine and equipment shall be examined for design, construction, materials, components, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

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4.5 Performance and accuracy test. A piece of annealed sheet steel having a width equal to the effective working length of the rolls and of a thickness equal to the capacity of the machine, as indicated in table I, shall be formed into a tubular shape, having an inside diameter of two inches larger than the roll diameter. For machines with dual capacity ratings, the shaped cylinder diameter shall be two inches larger than the roll diameter for the lighter rated material after pre-bending the edge of the sheet from the rear of the machine, then rolling in a normal fashion, only two passes through the machine shall be used in forming the tube. The inside diameter of the formed tube shall not deviate from a true cylinder by more than 1/16 inch, and shall not taper in size by more than 1/16 inch.

4.6 Examination of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Preservation, packaging, and packing. Preservation, packaging and packing shall be level A, B, or C as specified (see 6.2).

5.1.1 *Level A, B, or C.* The machines shall be preserved, packaged, and packed in accordance with MIL-P-18058.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. 123.

5.2.2 *Military activities.* Unless otherwise specified (see 6.2), marking shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use. Forming machines covered by this specification are intended for rolling sheet metals into cylindrical shapes and are designed for manual operation.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Style of forming machine required (see 1.2).
(*c*) Preproduction sample, if different (see 3.2).
(*d*) Special safety devices, if required (see 3.3.1).
(*e*) Painting, if different (see 3.4.5).
(*f*) Size number of machines required (see 3.6, 3.7 and table I).
(*g*) State when hardened and finished rolls are required (see 3.6.4).
(*h*) State type of main roll bearings required, if different (see 3.6.4.2).
(*i*) State when wire grooves are not required (see 3.6.7).
(*j*) State when demountable floor stand or pedestal is required (see 3.7.1).
(*k*) Optional equipment, if required (see 3.8.2).
(*l*) Nameplate, if different (see 3.9).
(*m*) Identification plate, if required (see 3.10).
(*n*) Handbook, if different (see 3.11).
(*o*) Anchor bolt plans, if different (see 3.12).
(*p*) Preproduction inspection, if different (see 4.2).
(*q*) Quality conformance inspection, if different (see 4.3).
(*r*) Levels of preservation, packaging, and packing required (see 5.1).
(*s*) Marking, if different (see 5.2.2).

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6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Machines, not otherwise indexed by name, S. U.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Machines, not otherwise indexed, S.U.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—WC

Navy—Ships

Air Force—WRA

Reviewer:

Army—WC, MO

Navy—Ships

Air Force—84

User:

Navy—WP, YD

Preparing Activity

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